Xuan Lin

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Education

University of Toronto, Bachelor of Applied Science
Computer Engineering **Expected Graduation in 2025**

Relevant Courses: Operating System, Algorithm and Data Structure, Computer Organization, Digital Systems, Programming Fundamentals (C/C++), Computer Architecture, Computer Security, Compilers and Interpreters, Database, Robotic Control and Modelling

Technical Skills

- **Programming skills**: C, C++, Python, MATLAB, SQL, Verilog, Assembly, Shell Scripting, K8S, HTML
- **Simulation**: Modelsim, NI Multisim, Quartus Prime, Wireshark, Simulink

Project & Academic Experience

Developer, Gesture-Controlled Virtual Instrument Interface, U of T **Sep 2024 - Apr 2025**

- Developed a gesture recognition module to classify hand gestures from video input using a machine learning model.
- Built the **Graphics module** to capture, preprocess, and compress real-time video input for gesture analysis using MediaPipe-compatible formats.
- Integrated gesture results into the GUI with visual overlays (gesture type, hand position, grid layout).
- Achieved real-time performance through efficient data handling and inter-module communication, ensuring low-latency interaction and robust system feedback.

Assistant Engineer, Serverless Distributed Systems, Huawei Canada **May 2023 – Aug 2024**

- Development of the adapter component for the URPC RDMA network
- Performance tuning for the adapter component
- Collaboration in the development of the data streaming framework
- Performance analysis for the data streaming benchmark

C++ Developer, City Map Software Design, U of T **Jan 2022 - Apr 2022**

- Built a street map software based on information from OpenStreetMap DatabaseCreated data structures for better querying within large-scale dataset and precomputed data beforehand to reduce repetitive coputation
- Implementing route-finding functions using classic algorithms like Dijkstra's algorithm, A*.
- Designed features and corresponding APIs to improve usability and responsiveness
- Developed GUI and visualized street information by combining graphics library (Gtk) and APIs previously wrote

C Developer, Game Project, U of T **Jan 2022 - Apr 2022**

- Designed a simple PvP board Strategy Game using C on a development board
- Implemented the game visualization by manipulating the pixel buffer
- Implemented the game interaction via interrupt signals and hardware keys on the board